Table 2: Summary of equilibrium types and their stability conditions in the attention-quality dynamics.

Type of equilibrium	Fixed point (x^*, y^*)	Stability condition
Collapse equilibrium	$(0, 0)$	Always stable
Boundary equilibrium	$\left(\frac{rn\sigma}{c_H}, 1\right)$	$\begin{cases} rn\sigma < c_H < rn, \\ \frac{c_A}{m\sigma(b_H - b_L)} + \frac{rn\sigma}{c_H} < 1 \end{cases}$
Coexistence equilibrium	$\left(1 - \frac{c_A}{m\sigma(b_H - b_L)}, \frac{c_H}{rn\sigma}\left(1 - \frac{c_A}{m\sigma(b_H - b_L)}\right)\right)$	$\begin{cases} m\sigma(1-\sigma)(b_H - b_L) > c_A, \\ \frac{c_A}{m\sigma(b_H - b_L)} + \frac{rn\sigma}{c_H} > 1 \end{cases}$

To clarify how attention constraints shape the coevolution of content quality and viewing behavior, we analyze the qualitative behavior of this replicator system. We first characterize all equilibria and derive the parameter thresholds that determine their stability. We then perform numerical integration of the system to examine trajectories and basins of attraction, illustrating how the system responds to changes in attention capacity and incentive strength.

Across these analyses, the dynamics organize into three characteristic regimes that govern long-run outcomes of attention and quality. In the collapse regime, both selective attention and high-effort provision vanish, and the system converges to a low-quality, non-selective state. In the boundary regime, consumers remain selective but providers exert only intermediate effort, leading to partially improved yet fragile information environments. In the coexistence regime, high-effort providers and selective consumers persist together with damped oscillations, maintaining quality through mutual adaptation. In the following sections, we characterize the parameter conditions under which each regime arises and analyze their dynamical and policy implications.

Analytical characterization of stable equilibria

We begin by analyzing the stability of the three equilibria identified in the replicator system: (i) the collapse equilibrium at the origin, (ii) the boundary equilibrium on $y = 1$ with $0 < x^* < 1$, and (iii) an interior coexistence equilibrium in which both high-effort providers and active consumers persist. As summarized in Table 2, the stability of these equilibria is governed by the interplay between attention capacity, the cost of active consumption, and the effective reward for high-effort provision.

The origin $(x, y) = (0, 0)$ is always locally stable, representing a collapse regime where both selective attention and high-effort provision disappear. In particular, this equilibrium becomes the only attractor when the following condition holds:

$$m\sigma(1 - \sigma)(b_H - b_L) < c_A. \tag{1}$$

The derivation of this threshold is provided in the SI Appendix. Intuitively, the term $\sigma(1-\sigma)$ indicates that discrimination is weakest when attention is either too scarce or too dispersed, making selective viewing unprofitable and preventing high-quality content from gaining traction. In this parameter region, no nontrivial equilibria exist, and all trajectories converge to the origin regardless of initial conditions. In such environments, consumers find it unprofitable to engage in selective attention, and low-effort content dominates in the steady state.

In contrast, when $m\sigma(1 - \sigma)(b_H - b_L) > c_A$, nontrivial equilibria become possible. Which equilibrium stabilizes depends on the ratio of the effective reward to the production cost, $rn\sigma/c_H$. If

$$\frac{c_A}{m\sigma(b_H - b_L)} + \frac{rn\sigma}{c_H} > 1, \tag{2}$$

the interior coexistence equilibrium is locally stable. In this regime, both high-effort providers and active consumers persist at positive levels, sustaining quality endogenously through mutual reinforcement. In contrast, if the opposite inequality holds,

$$\frac{c_A}{m\sigma(b_H - b_L)} + \frac{rn\sigma}{c_H} < 1, \tag{3}$$

the boundary equilibrium becomes stable: all consumers remain active ($y \to 1$), but providers exert only intermediate effort. This state represents a partially functional but not fully self-sustaining information environment.

These analytical results indicate two hierarchical thresholds that shape the sustainability of quality under limited attention. The first threshold, $m\sigma(1 - \sigma)(b_H - b_L) = c_A$, separates collapsed from non-collapsed systems and depends solely on consumers' ability to distinguish between high- and low-quality content. The second threshold, determined

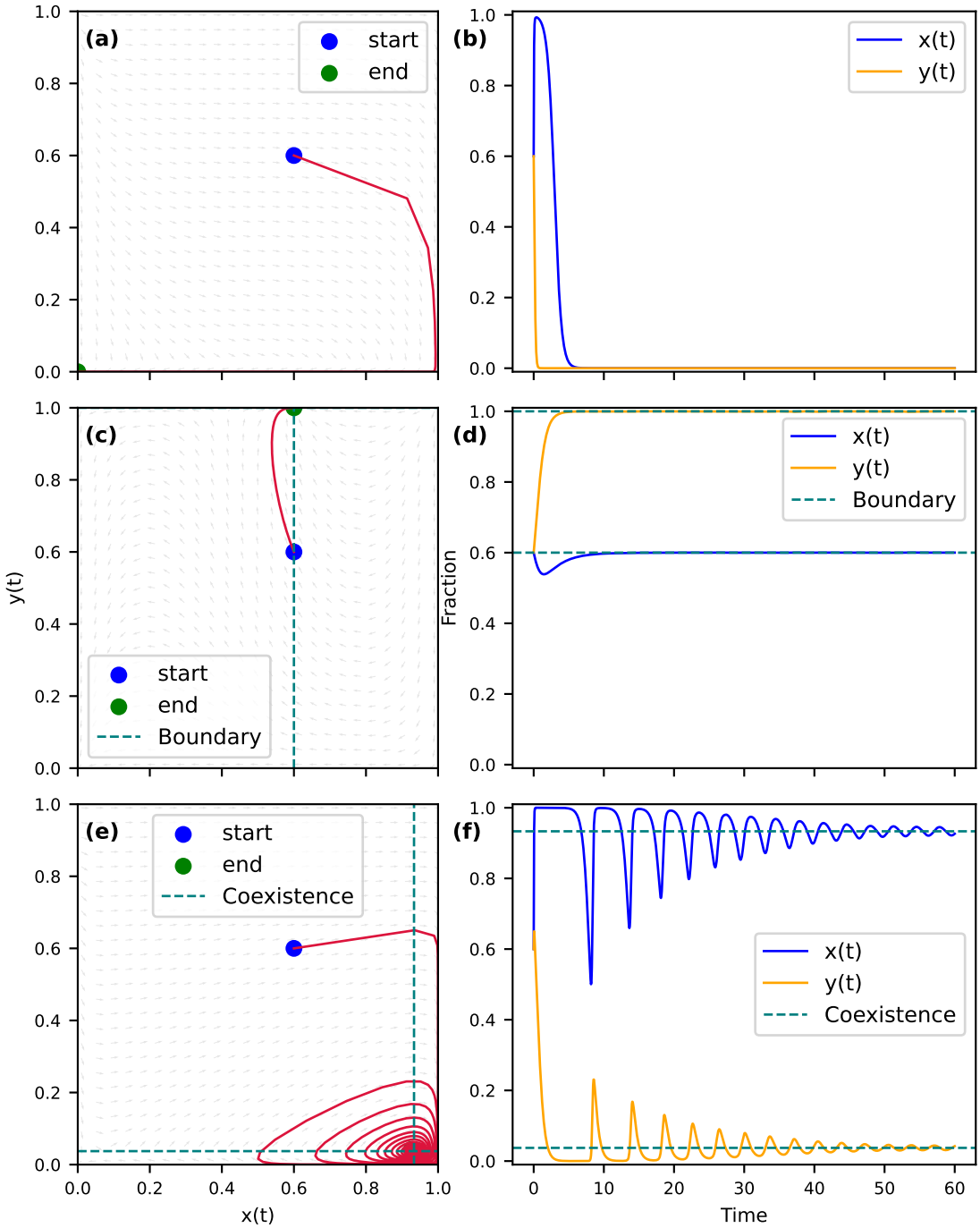

Figure 2: Representative trajectories of the replicator dynamics under three parameter regimes. Parameters were chosen to satisfy the corresponding stability conditions: (a,b) Collapse: $m = 20$, $n = 100$, $\sigma = 0.5$, $r = 1$, $b_H - b_L = 1$, $c_A = 10$, $c_H = 2$. (c,d) Boundary: $m = 1$, $n = 10$, $\sigma = 0.5$, $r = 0.12$, $b_H - b_L = 10$, $c_A = 1$, $c_H = 1$. (e,f) Coexistence: $m = 20$, $n = 100$, $\sigma = 0.5$, $r = 1$, $b_H - b_L = 3$, $c_A = 2$, $c_H = 2$. The green dashed lines mark the equilibrium.

by the balance between reward and cost parameters, divides the non-collapsed region into boundary and coexistence regimes. Together, they show that maintaining high-quality information requires not only sufficient discriminability among consumers but also an adequate incentive structure for providers.

Dynamic behavior across three regimes

Next, we examine how system trajectories approach these equilibria under varying parameter regimes. Figure 2 presents representative vector fields and time evolutions for the three regimes: collapse, boundary, and coexistence. These trajectories were generated by numerically integrating the replicator equations with a standard adaptive Runge–Kutta method (RK45), using parameter values chosen to satisfy the stability conditions of each regime.

When the active-consumption cost c_A is high or the quality gap $b_H - b_L$ is small, consumers lack sufficient motivation to search for high-quality content. Selective viewing fails to emerge ($y \to 0$), and providers lose incentives to exert high effort ($x \to 0$), leading both populations to a state dominated by low-quality content and non-selective behavior.

In the boundary regime, consumers sustain selective attention ($y \to 1$), but the effort level of providers remains moderate, with x converging to an intermediate value. This regime represents an environment where demand for quality exists, yet the reward for high-effort provision is inadequate to sustain full-quality production.

In the coexistence regime, both providers and consumers persist at positive equilibrium levels ($x > 0, y > 0$), yet convergence toward this steady state is not monotonic. The trajectories exhibit mild oscillations around the equilibrium, reflecting feedback between supply and selection. When the share of low-quality content increases, consumers become more selective (y rises), which restores incentives for high-effort creation and raises x. As high-quality content becomes abundant, the need for active searching decreases, and y gradually falls. This in turn reduces x, generating a slow cyclical adjustment. Because these adjustments weaken as the system approaches the fixed point, the feedback loses strength over time, producing damped rather than persistent oscillations.

This damped cycle reflects adaptive consumer behavior: selectivity increases only when necessary, allowing consumers to minimize active-consumption costs while supporting the persistence of high-quality supply. Thus, the coexistence regime represents a dynamically maintained state in which high-effort provision and selective attention reinforce one another.

Basins of attraction across regimes

Having examined the typical trajectories within each regime, we next investigate how their basins of attraction vary across parameter settings. The size and structure of these basins differ substantially between the boundary and coexistence regimes.

Figure 3 shows the final equilibrium reached from systematically varied initial conditions (x_0, y_0). Red regions denote initial conditions that converge to the coexistence or boundary regime, while gray regions correspond to trajectories converging to the collapse regime. Arrows in the background indicate the vector field of the replicator dynamics.

When the boundary equilibrium is stable (left panel), the region of initial conditions leading to coexistence is limited. Without a sufficiently large initial fraction of high-effort providers or active consumers, the system collapses to $(x, y) \to (0, 0)$. This indicates that the boundary regime is highly sensitive to initial conditions: quality and selectivity must already be present at appreciable levels for the system to avoid degeneration.

In contrast, when the internal coexistence equilibrium is stable (right panel), the basin of attraction expands substantially. Even when both x and y start from low levels, the feedback between effort and attention drives the system back toward coexistence. This result indicates that the coexistence regime is robust across a wide range of initial conditions, whereas the boundary regime emerges only when selective attention or high-effort provision is already prevalent.

Determinants of collapse and policy implications

The key bifurcation of the system separates the collapse regime, where both strategies vanish, from the non-collapsed regimes, where at least partial coexistence of high- and low-quality content persists. This transition primarily depends on consumers' ability to discriminate between quality levels, which is characterized by $m\sigma(1-\sigma)(b_H - b_L) = c_A$. Here, $\sigma(1-\sigma)$ captures a nonlinear effect of attention capacity: discrimination weakens when attention is either too scarce or too dispersed. If this balance between discriminability and active-consumption cost fails to exceed this threshold, selective engagement does not emerge, and high effort cannot be sustained. In this case, the system collapses regardless of initial conditions.